

UNG FACILITIES PLANNING & GIS

Version 4.0

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Analyzes GIS room number quality — gaps, duplicates, wall polygons, floor counts

RoomNumberAnalysis_V4.0.py

Purpose

Analyzes room_num values across GIS floor plan feature classes to identify data quality issues: duplicate room numbers (critical — remediate immediately), sequence gaps, wall polygon sanity (one 999N polygon per floor expected), and floor count mismatches against the bldg_levels domain. Supports single building, single floor, or entire campus analysis modes. Output is a console summary plus an Excel workbook with four sheets.

When to Run

- On demand — not part of the weekly pipeline
- During room numbering remediation campaigns
- Before or after any floor plan modification that touches room numbers
- Periodically as a GIS data quality audit

Requirements

- ArcGIS Pro / arcpy
- Read access to UNG_ENTERPRISE_UPDATED_WORKING.gdb

Analysis Modes

Mode 1 — Single building: Analyzes all floors of one building

Mode 2 — Single floor: Analyzes one floor within one building

Mode 3 — Entire campus: Analyzes all buildings in one campus feature class

What It Checks

- Duplicate room numbers within a building+floor group — Critical triage, remediate immediately
- Sequence gaps — missing room numbers between the lowest and highest used numbers
- Next available room number per floor
- Wall polygon sanity — expects exactly one 999N polygon per floor
- Floor count vs bldg_levels domain — flags mismatches between floors found and declared level count

Outputs

Item	Value	Notes
[Building]_RoomAnalysis_TIME STAMP.xlsx	Scripts\RoomNumbering\Analysis_Files\	Four sheets: Summary, Duplicates, Gaps Only, Detail
Console summary	Printed to screen	Key counts and critical flags

Triage Priority

Critical: Duplicate room numbers — remediate before next pipeline run; duplicates cause the updater to skip those polygons entirely (ambiguous write risk)

Functional: Floor count mismatches and wall polygon issues

Cosmetic: Sequence gaps — document and monitor but do not force renumbering

Room Numbering Policy

GIS is the authoritative record for room numbers. Banner is maintained to match GIS. The A/B suffix convention applies to split rooms. Any room number that goes into use must be reported to Facilities before use. Banner space inventory access must be a stated job requirement for the successor to this position.